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6.1 Extrapolating Price from Volatility

Estimating Future Prices with VIX

In Level I, we explained that VIX® measures the 30-day implied volatility of the S&P 500® (SPX) Index using a calculation based on near- and next-term S&P 500 put and call options. This 30-day implied volatility is expressed as an annualized number. Details of this calculation can be found at [Cboe.com](https://www.cboe.com).¹

Estimates of quarterly, monthly, weekly, and daily standard deviations of prices can be calculated from the daily values of VIX, which, again, is the 30-day implied volatility expressed as an annualized number.

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6.1 Extrapolating Price from Volatility



FIGURE 6.1.1 S&P 500 index (top), VIX index (bottom)

At the end of the first quarter of 2024, VIX, which is usually expressed as a percentage, was 13.01%. We can now estimate volatility, in this case standard deviation, for different periods of time using the following formulas:

$$\text{Quarterly volatility} \approx \frac{\sigma}{2 \text{ or } \sqrt{4}}$$

$$\text{Monthly volatility} \approx \frac{\sigma}{3.46 \text{ or } \sqrt{12}}$$

$$\text{Weekly volatility} \approx \frac{\sigma}{7.21 \text{ or } \sqrt{52}}$$

$$\text{Daily volatility} \approx \frac{\sigma}{15.87 \text{ or } \sqrt{252}}$$

where σ (sigma) = annualized volatility, assuming 252 trading days in a year

Monthly volatility is calculated by dividing 13.01%, the current VIX reading, by 3.46 or the square root of 12.

$$\text{Monthly volatility} \approx \frac{\left(\frac{13.01\%}{100}\right)}{3.46 \text{ or } \sqrt{12}} \approx \left(.0376 * 100\right) \approx 3.76\%$$

This predicts a 68.2% (one standard deviation or sigma) probability of prices being between +3.76% and -3.76% from the current price in 30 days' time. Using the SPX closing price of 5,254.35 on March 28, 2024 from Figure 6.1.1, we can now

Implied 30-day price move $\approx$ current SPX index value $\times$ VIX monthly volatility

$$\text{Implied 30-day move} \approx 5,254.35 \times (3.76\% / 100)$$

$$\text{Implied 30-day move} \approx \pm 197.56$$

Finally, add and subtract the value of the implied 30-day move to conclude, with a 68.2% probability over the next 30 days, the options market is expecting SPX to trade between 5,451.91 and 5,056.76.

The standard deviation calculated from implied volatility also gives market participants an opportunity to estimate daily and weekly trading ranges.

Daily volatility is calculated by dividing 13.01%, the current VIX reading, by 15.87 or the square root of 252.

$$\text{daily volatility} \approx \frac{\left(\frac{13.01\%}{100}\right)}{15.87 \text{ or } \sqrt{252}} \approx \left(.0082 \times 100\right) \approx 0.82\%$$

This predicts a 68.2% (one standard deviation or sigma) probability of prices being between +0.82% and -0.82% from the current price in one day's time. Now calculate the implied 30-day move in SPX and add and subtract that from the current price.

Implied single-day move $\approx$ SPX index value $\times$ VIX daily volatility

$$\text{Implied single-day move} \approx 5,254.35 \times (0.82\% / 100)$$

$$\text{Implied single-day move} \approx \pm 43.09$$

Using the SPX closing price of 5,254.35 on March 28, 2024 from Figure 6.1.1, the options market expects price to trade in the range of 5,297.44 and 5,211.26, with a 68.2% probability over the next day.

Estimates of future prices calculated in this manner can be a useful tool for traders to plan their strategies. The upper and lower levels could be used to sell or buy in conjunction with other technical analysis applications including establishing strike prices for writing covered calls or buying puts for downside protection against losses.

¹ Cboe. "Volatility Index Methodology: Cboe Volatility Index." November 20, 2023.
https://cdn.cboe.com/api/global/us_indices/governance/Volatility_Index_Methodology_Cboe_Volat



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Estimating Future Prices with VIX



6.1 Extrapolating Price from Volatility

Estimating Future Prices with VIX

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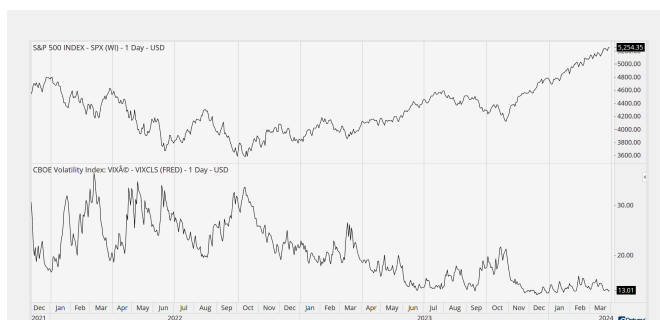


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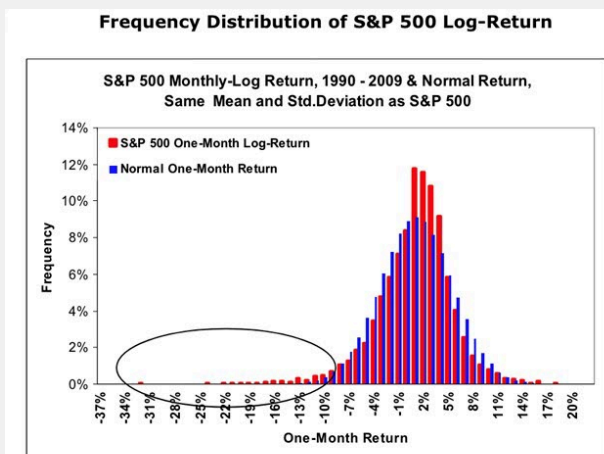
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6.1 Extrapolating Price from Volatility

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Limitations of VIX-Based Forecasts

In applying VIX to estimate future prices, readers would do well to keep in mind that the standard deviation in this instance assumes a normal distribution. Studies on one month S&P 500 return distributions, as shown in Figure 6.1.2, show that price actually exhibits a nonnormal distribution. The distribution has a positive kurtosis, meaning longer tails, especially to the left in this case, and a taller or more “peaked” distribution.



Source: CBOE

FIGURE 6.1.2 Chart showing S&P 500 returns are not normally distributed

Source: Cboe

<https://cdn.cboe.com/resources/indices/documents/SKEWwhitepaperja>

Longer tails mean more outliers. This speaks to the need for effective risk management. Also, the more “peaked” or taller shape of the S&P 500 return distribution implies that VIX sometimes underestimates upside and downside price movement in the S&P 500. This tendency of underestimating forward returns in the S&P 500 illuminates its limitation.

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